What happens when predictions are disconfirmed?
Disentangling the underlying processes by using an implicit memory test

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Studies have shown that prediction during language comprehension entails the pre-activation of the expected word and its semantic and form-related neighbors (Federmeier & Kutas, 1999; Huettig et al., 2022; Nieuwland, 2019). But what happens with these pre-activated representations when predictions are disconfirmed? Previous studies have mainly investigated this question by means of explicit recognition memory tests. The results demonstrated false remembering of expected words and their semantic neighbors which suggest that expected words and their semantic neighbors remain active in memory even when predictions are disconfirmed (Haeuser & Kray, in press; Hubbard & Federmeier, 2024; Hubbard et al., 2019). However, in those tests participants make an overt memory judgment: Irrespective of any prediction, participants may simply remember the sentence context that fits the word being judged. Consequently, the results of these studies do not necessarily reflect prediction effects (Haeuser & Kray, in press). In addition, previous work has suggested that prediction suppresses form-related neighbors (e.g., leaf when leader is predicted, Haeuser & Borovsky, 2024). Therefore, the aims for the present study were twofold: 1. Is there a downstream effect of disconfirmed predictions in a paradigm that does not measure overt memory retrieval? 2. Do form-related neighbors get inhibited after prediction disconfirmations? We investigated this question by means of an implicit memory test (e.g., Ren et al., 2024) in which participants were asked to quickly recognize blurred words that became gradually deblurred over time (Figure 1). Faster recognition times indicated that words were pre-activated in memory.

Native German-speaking adults (n = 77, m = 24, f = 52, nb = 1, *M* = 22.9 years old, range = 18-35 years old) read highly constraining German sentences which ended in an expected or an unexpected word in a self-paced reading task (Table 1). After a 5-minute distractor task, participants performed the implicit memory test. If predictions are maintained in memory, we expect that initially predicting a word leads to its pre-activation and of its semantic neighbors and results in faster recognition times for expected words compared to their semantic neighbors. If form-related neighbors get inhibited, we expect that form-related neighbors are recognized more slowly than expected words.

Preliminary results (Table 2, Figure 2) showed that words were recognized faster when the prediction has been confirmed than when the prediction has been disconfirmed suggesting an overall effect of prediction on the recognition times. However, expected and semantically-related words were recognized equally quickly, which challenges the interpretation of a pure prediction effect. In line with our hypothesis, form-related words were recognized more slowly compared to expected words.

In sum, regarding our first question, whether there is a prediction effect, the results are inconclusive. On the one hand, we found that recognition times are faster after prediction confirmations, which indicates that expected words and their related neighbors are maintained in memory. In the case of prediction disconfirmations, their memory representations are only weakly maintained probably to ensure successful processing of the word presented instead. On the other hand, we did not find a recognition time advantage for expected words over their semantic neighbors, which would have constituted straightforward evidence that prediction engendered the observed effects. Instead, the effect might have been confounded by a context association effect as both words were related to the sentence context (Haeuser & Kray, in press). However, regarding our second question, we found a clear difference between expected and form-related words which cannot be explained by a pure context association effect. Future studies need to further disentangle downstream prediction effects and context association effects.

Figure 1

Example of the gradually deblurred effect on the word "Tisch" (table) over time

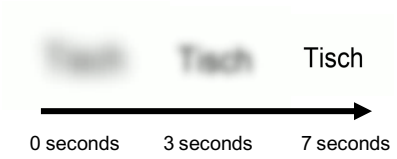

Figure 2

Fitted log-transformed recognition times for the different word conditions

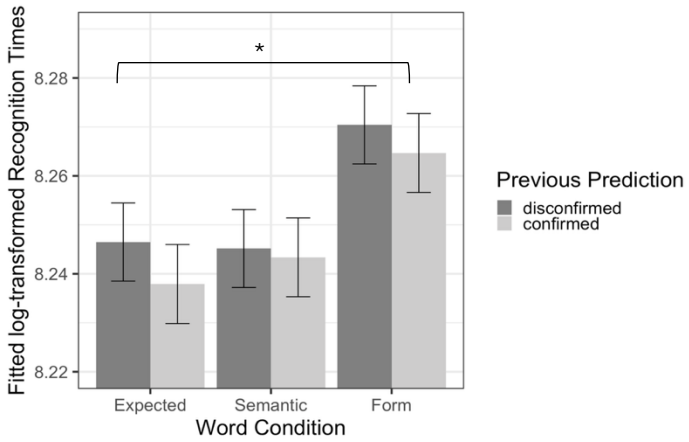

Table 1

Example sentence and the corresponding words presented in the implicit memory test

Sentence	Prediction (dis)confirmed?	Old Words	Lure Words		
		Confirmed expected word	Disconfirmed expected word	Semantically-related word	Form-related word
Wenn Gäste zum Kaffee kommen, deckt Frau Meier immer sehr liebevoll den Balkon im Haus. *When guests come over for coffee, Mrs. Meier always sets the balcony in the house with great care.*	Prediction disconfirmed	/	Tisch *table*	Stuhl *chair*	Tischler *carpenter*
Wenn Gäste zum Kaffee kommen, deckt Frau Meier immer sehr liebevoll den Tisch im Haus. *When guests come over for coffee, Mrs. Meier always sets the table in the house with great care.*	Prediction confirmed	Tisch *table*	/	Stuhl *chair*	Tischler *carpenter*

Table 2

Model syntax and output (significant results in bold)

log (recognition time in ms) ~ word condition_sum_coded *
(dis)confirmed_treatment_coded
+ scale (trial number in implicit memory test)
+ scale (word frequency)
+ scale (word length)
+ scale (number of form-related neighbors)
+ scale (visual acuity)
+ scale (cloze probability of the corresponding sentence)
+ scale (trial number in self-paced reading task)
+ (1 | participants) + (1 + (dis)confirmed | words)

(3 levels of word condition: expected, semantic, form;
2 levels (dis)confirmed: confirmed, disconfirmed)

	b	SE	*t*	*p*
(Intercept)	**8.254**	**0.006**	**1391.477**	**< .001**
expected_semantic	0.001	0.009	0.144	.886
expected_form	**-0.024**	**0.009**	**-2.588**	**.011**
(dis)confirmed	**-0.005**	**0.002**	**-3.202**	**.002**
trial number in implicit memory test	-0.002	0.002	-0.953	.342
word frequency	-0.006	0.004	-1.556	.122
word length	**0.016**	**0.005**	**3.088**	**.003**
number of form-related neighbors	-0.001	0.005	-0.244	.808
visual acuity	**-0.010**	**0.005**	**-2.105**	**.039**
cloze probability of the corresponding sentence	-0.002	0.004	0.488	.626
trial number in self-paced reading task	-0.005	0.004	-1.280	.203
expected_semantic * (dis)confirmed	-0.007	0.004	-1.656	.101
expected_form * (dis)confirmed	-0.003	0.004	-0.687	.493

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